

for this category is 75 per cent with speed tests getting 60 per cent weightage and quality tests getting 15 per cent.

3D speed tests

Our speed benchmarks were broken down into OpenGL and Direct3D as every 3D game released today supports either one of these standards. Therefore it's imperative that your graphics card performs well in these tests.

■ **Quake III Arena (OpenGL):**

Extremely bandwidth-sensitive, the *Quake III Arena* engine still has what it takes to bring most graphics chips down to its knees. We used an in-game pre-recorded demo (demo four) with the retail version of *Quake III Arena* patched to version 1.30. We ran the tests in three different modes (resolution x colour depth x texture depth): Normal (640x480x16x16), High (800x600x32x32), and Maximum (1024x768x32x32)—and the average fps over three test runs was noted down.

■ **3DMark2001 SE (Direct3D):**

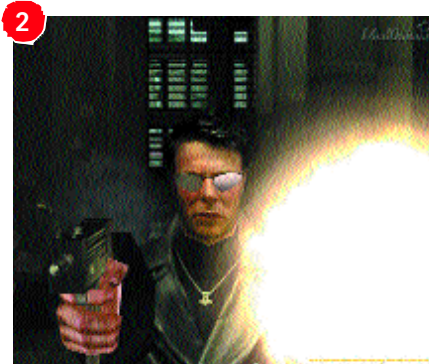
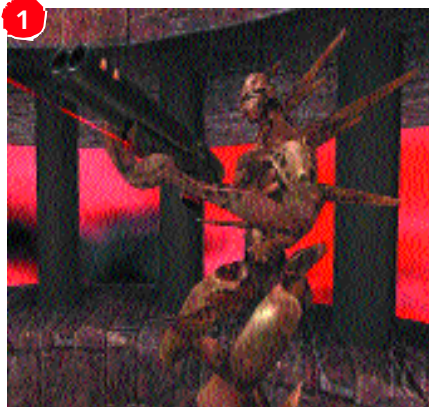
3DMark is a synthetic benchmark that runs 21 tests to evaluate the graphics performance of your system and brings out any bottlenecks that might be hindering its performance. The test suite evaluates various features supported by the graphics chip including hardware Transform and Lighting, fill rate, image quality, etc. This new version also tests the performance of the latest programmable graphics chips. We ran 3DMark at the default settings (1024x768 at 32-bit colour with trilinear filtering and hardware T&L support enabled).

■ **Comanche 4 (Direct3D):**

Comanche 4 is a helicopter-based action game that pushes the video chipsets to their limits. Good performance under this benchmark guarantees a safe and future-proof buy. We ran the tests in three different settings: 640x480x16, 800x600x16 and 1024x768x32—and the average fps was noted down. A chipset has to support Hardware T&L for this benchmark to run.

2D speed tests

For the 2D speed test, we used PCMark2002 Pro to check the speed of a particular chipset in drawing 2D lines, curves and images. This program runs a series of benchmarks that tells you whether your graphics chip can handle the day to day work under the latest operating systems. This test was performed under Windows XP.



Our test suite consisted of two games and one benchmark application:

1. *Quake III Arena*, which is an OpenGL-based 3D game
2. *3DMark2001 SE*, which is a benchmark that tests DirectX 8.1 compatibility of a video chipset
3. *Comanche 4*, which is a Direct3D-based game that utilises advanced pixel and shader effects present in newer graphics chipsets

Quality tests

The 2D image quality of the graphics chips was tested under Windows XP. We used an ordinary word document filled with text in different font types and sizes to evaluate the 2D quality of the graphics chips. The 3D image quality was tested using *Quake III Arena* with FSAA and anisotropic filtering turned on (1024x768x32, 2x/4x FSAA and anisotropic filtering). We took screenshots at selected points in the game and checked for quality issues such as aliasing problems, texture artefacting and colour bleeding.

Value for money

This provides us with a price-performance ratio. A chipset offering good performance at a great price will score higher in this category than one offering excellent performance with a hefty price-tag. We



computed this value by taking into consideration the price of the cheapest card available in the market for each graphics chipset. The weightage allotted to this category was 25 per cent.

How they fared

Sure games are cool, games are fun, but not if they run like a slideshow on your system. To truly enjoy a game, a decent frame rate is an absolute essential. Of course, the word decent is relative and can mean anything from 30 fps for a role-playing or an adventure game, to 60 fps for an action packed game like *Quake III Arena*.

3D performance

The main aim here was to evaluate which chipset was the ultimate performer on the basis of raw speed. In 3D graphics terminology, speed is measured in frames per second (fps) or the number of frames that a chipset can output in one second.

If you want an enjoyable experience, 30 fps is the minimum that you should expect from any game. However, games are fickle things—in one scene you might see trees waving in the air and nothing but some rocks to keep you company, while the next second the entire screen could be filled with monsters and general baddies of all sizes and shapes. A scenario like this is not uncommon, regardless of the type of game and it will always cause a sudden and drastic drop in the frame rate. Thus, it is not without reason that we have considered a frame rate of 60 fps as the minimum that your graphics chipset should run a game at. This will be our cut-off value, anything above is cool and anything below is not.

Quake III Arena (OpenGL)

With the *Quake III Arena* engine powering most of the new first person action games, you're assured of adequate performance across the board if your graphics chip can handle this test.